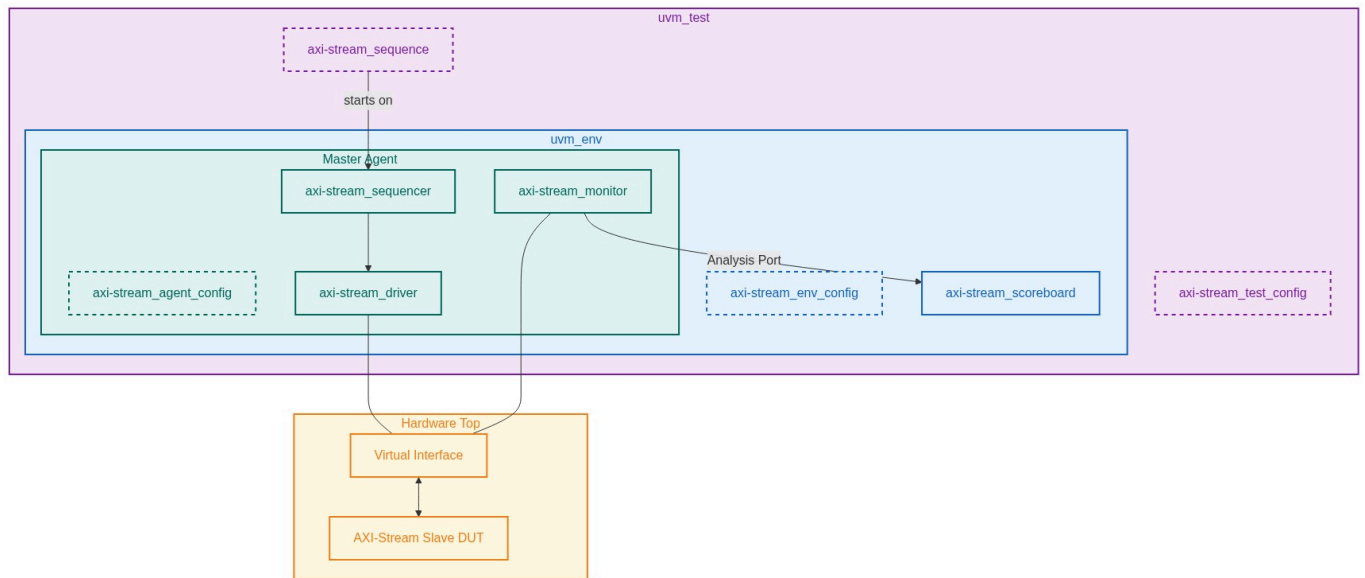


Verification Plan Report

Project: AXI-Stream Slave DUT

Version: 3.1

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Ordering (OoO)	Strictly Ordered Stream	Reordering is prohibited. Simplifies scoreboard logic as transfers must arrive in dispatch order.
Packet Structure	Sparse Stream Support	Enables TKEEP/TSTRB qualifiers for null byte handling. Requires byte-level scoreboard tracking.
Interleaving	Multi-Stream Interleaving	Supported via TID signals. Verification must ensure TID-to-TID switching does not corrupt packet boundaries.
AXI5 Features	Parity & Wakeup	Adds TWAKEUP and TDATACHK. Increases verification scope to include fault injection and low-power states.

3. Test Requirements

Index	Name	Description
REQ_HANDSHAK E_01	TVALID/TREADY Handshake Rules	Verification of the 2-way flow control. TVALID must not wait for TREADY. TVALID must remain stable until TREADY is asserted.

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REQ_BYTE_QUAL_01	TKEEP and TSTRB Handling	Correct identification of valid bytes (TKEEP) and data vs position bytes (TSTRB).
REQ_PARITY_01	Odd Parity Protection	Identify single-bit flips on TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, and TUSER.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_RAND_PKT_SIZE	Send packets with randomized beat counts and TREADY backpressure.	REQ_HANDSHAKE_01
TC_SPARSE_STREAM	Verify slave can handle sparse streams with intermittent TKEEP/TSTRB bits.	REQ_BYTE_QUAL_01
TC_PARITY_ERR	Inject odd parity errors on TDATACHK and verify error signal detection.	REQ_PARITY_01

5. Functional Coverage Groups

Group Name	Description
cg_handshake	Coverage for TVALID/TREADY being HIGH/LOW simultaneously.
cg_data_width	Ensure all data bits in TDATA have toggled.

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	In	1	Clock
TVALID	In	1	Transmit valid
TREADY	Out	1	Receive ready
TDATA	In	DATA_WIDTH	Payload
TDATACHK	In	DATA_WIDTH/ 8	Parity bits (AXI5)

7. Verification Environment Structure

Root: axi_strm_slv_env/

|-- agent : Master driving Slave

|-- env : UVM Environment